Forging the Future: Market Intelligence and Growth Strategy for High-Performance Metals Through 2035

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6 December 2026

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1. Introduction

The decision this report asks the board to take is a ranking of qualified orders, not a capacity addition sized to a mixed trade code. The three priorities are aerospace billet, approved sheet and repair wire, corrosion resistant plate and welding products where a named grade is imported or replaces a specified supplier, and precision electrical strip and wire only where the grade is one the mill already finishes. EU nickel flat apparent consumption of 57.8 million kilograms in 2024 is the ceiling of a mixed code that still joins pure nickel and nickel alloy, and it is not qualified alloy demand and not room under the Acerinox finish figure (Eurostat, 2025d). Flat products are a net export of that mixed code, so undifferentiated flat tonnes are refused, while bar, rod and wire is a logged net import that is still a rounded estimate and is a test of the existing book rather than a case to add bar capacity (Eurostat, 2025b; Team, 2026f).

The same ranking carries a stop list that is as specific as the pursue list. Glow plugs and the rest of the diesel only set come off the growth list, undifferentiated stainless is left where stainless already meets the duty, China implant tenders are not pursued after the recorded price cuts, and no electrolyser or powder programme is funded without a financed buyer. Oil and gas is kept only where an approval the company itself holds supports the margin, automotive is harvested only where the account is already profitable, and medical implant bar stays an option until an implant maker and a regulatory path are the company's own. The first 24 months test the European bar position against the company's bar book, cap the live qualification list, and release finishing capacity only against a repeat order that still clears the downside case.

Because those choices depend on keeping each metal at the stage the statistics measure, the chapters that follow separate supply, trade, industrial use and the path to 2035. World mine production of contained nickel was an estimated 3.90 million tonnes in 2025, while stainless melt shop output in the same year was 64.157 million tonnes of gross steel, and neither figure is a shipment of plate or bar (U.S. Geological Survey, 2026s; worldstainless, 2026a). Supply asks what commercial form is produced and by whom, trade asks where that form is imported and exported, demand asks which component and buyer turn the form into an order, and the outlook tests whether that order still exists when no tonne share of the mixed code is invented. Münvest meets this system as a producer of qualified mill products, which is why the next chapter keeps the VDM Metals catalogue as an application benchmark only.

2. Report Objective and Scope

The objective follows from that separation of stages and identifies where changes in metal supply, trade and industrial demand create commercially relevant opportunities for Münvest

through 2035. The central question is how the production, availability, trade and industrial uses of selected metals shape opportunities in high performance materials through 2035.

The scope that answers the question covers ten named metals and a residual category, together with seven industries: automotive, oil and gas, energy and environmental technology, aerospace, chemical processing, electronics and electrical engineering, and medical technology. The metal chapters provide the production and form context, and the industry chapters keep the application evidence for those buyers and close each sector with what the company should do. VDM Metals' published portfolio is an application benchmark only, and it is not evidence of Münvest's plants, customers or share, while VDM and Haynes are counted once because both sit inside Acerinox.

Observation years stay attached to each figure, because mined metal, primary metal, mill product shipments and industry revenue have different boundaries and are not interchangeable measures of market size. No aggregate addressable market value is asserted where comparable data are unavailable. Apparent consumption, defined in the next chapter as sold production plus imports minus exports, is one of those non interchangeable measures and is applied only where all three legs are logged.

The scenarios in Chapter 7 are sensitivities for relevant application demand, and they are not forecasts of metal prices, company revenue or investment returns. Internal sales, margins, approvals, capacity and customer lists are required before those findings become a business plan, which is why Chapter 3 states how the four dimensions will be read together.

3. Introduction to the Analytical Framework and Methodology

Because those boundaries are easy to cross, the report uses one metal centred value chain with four dimensions: Supply, Trade, Demand and Outlook. Supply is assessed metal by metal in Chapter 4 through commercial forms, production stages, producer structure, availability and prices. Trade is assessed in Chapter 6 through product classifications, exporting and importing markets, and sourcing dependence. Demand is examined industry by industry in Chapter 5, linking drivers to components, materials, delivery forms and buyers. Outlook is assessed in Chapter 7 through explicit scenarios to 2035, and Chapter 8 integrates the four dimensions for Münvest. The sequence is metal first and industry second, so the reader sees what the material is before judging who buys it.

Table 1: How the four dimensions map onto the data chain

Dimension	Chapter	Data chain steps
Supply	Chapter 4	Material and product form
Trade	Chapter 6	Geography
Demand	Chapter 5	Application and industry
Outlook	Chapter 7	Market size and growth

Sources: The team framework stated in this chapter supplies every row, and the rows classify steps inside Supply, Trade, Demand and Outlook rather than drawing on an external dataset.

The mapping keeps each figure in the step it measures, so a mine statistic belongs to Material, a PRODCOM plate balance belongs to Product form, and a component and its buyer belong to Application and Industry. A corridor belongs to Geography, and a 2035 sensitivity belongs to Market size and Growth, while prices stay inside each metal market because a commodity benchmark is not the selling price of a qualified alloy.

Apparent consumption is the mill product check inside Supply and Trade, and for one product code, one area and one year it equals sold production plus imports minus exports, with the same identity available in euro where all three value series exist (Team, 2026d). The result records the trade balance of that code, while end use alloy demand and stainless melt tonnage remain separate series. Chapter 4 applies the identity to EU and German nickel flat products and to EU nickel bar, and Chapter 6 compares those results with Comtrade, with named mill capacity and with stainless melt. Where any leg is missing, no apparent consumption figure is stated in the text.

Sources stay at the stage they measure. Commodity definitions and mine statistics follow the United States Geological Survey Mineral Commodity Summaries, and company filings and industry sources support applications and competitive structure. Two figures are not treated as comparable merely because both name the same metal. Contained metal stays distinct from gross shipment weight, mine output from refined output, primary metal from scrap, actual output from capacity, and group output from an ownership share. Missing observations stay missing rather than being filled from a neighbouring series. Geographic concentration is calculated separately for extraction and for processing.

Chapter 8 reads the four dimensions against eight assessment criteria: market attractiveness, growth to 2035, strategic fit, competitive intensity, technology attractiveness, customer attractiveness, entry barriers and differentiation potential. The project model holds the weights, and this text does not collapse the criteria into one score, because shipment shares by application were not supplied. Coverage is deeper for nickel and cobalt alloys and for selected corrosion resistant materials than for every metal in the scope, and the later profiles mark that

limit instead of filling it with an estimated total. The industry chapters then take the forms that survive this screen and ask who buys them.

4. Metal Markets: Production & Form

The metal profiles apply the measurement rule from Chapter 3 one material at a time. Each profile states the commercial form a figure refers to, because ore, intermediate, refined metal and mill product answer different sourcing questions. Country rankings appear only where a dated dataset is in the source log. Nickel is first, because Table 2 is the only scale exhibit and the later metals are then set against that basket rather than against a repeated ladder of the same tonnes.

4.1 Nickel

Nickel is the clearest bridge from the upstream metal balance to the mill products Münvest would sell, which is why the stages below are kept on separate lines before any commercial reading is taken. World mine production of contained nickel was 3.71 million tonnes in 2024 and an estimated 3.90 million tonnes in 2025, and Indonesia accounted for an estimated 2.60 million tonnes in 2025 (U.S. Geological Survey, 2026s, 2026t, 2026u). These mine figures are contained nickel in ore, and they become useful for sourcing only after primary metal, stainless melt and mill product are kept on separate lines.

Primary production is the next stage, covering ferronickel, nickel pig iron, cathode and related refined units. The International Nickel Study Group puts world primary production at 3.875 million tonnes in 2025 and usage at 3.577 million tonnes, with a stated surplus of 299 thousand tonnes, after a 2024 surplus of 116 thousand tonnes (International Nickel Study Group, 2026a, 2026b, 2026c). The October 2026 meetings put the 2026 surplus at 19 thousand tonnes and the 2027 surplus at 70 thousand tonnes, so the near term balance is a stated tonne surplus rather than a description of that surplus as small (International Nickel Study Group, 2026d). London Metal Exchange nickel is primary metal of at least 99.80 percent purity, which is a narrower pool than the headline primary total, and the 2025 annual average cash price was about USD 15,000 per tonne (London Metal Exchange, 2026; U.S. Geological Survey, 2026v).

Stainless melt is the main first use of primary nickel and is counted as gross steel. World melt shop production was 64.157 million tonnes in 2025, of which the European Union produced 5.659 million tonnes (worldstainless, 2026a, 2026b). That stainless system is measured in millions of tonnes of steel, while the mill forms below are measured in thousands of tonnes of plate, sheet, bar and wire, and Table 2 is the only place the full ladder is set out.

Table 2: Logged nickel mill product is tens of thousands of tonnes, and the flat code still mixes pure nickel with alloy

Stage	What is counted	Year	Quantity
World mine production	Contained nickel in ore	2025	3.90 million tonnes
World primary production	Primary nickel	2025	3.875 million tonnes
World primary use	Primary nickel	2025	3.577 million tonnes
World stainless melt	Gross steel	2025	64.157 million tonnes
EU stainless melt	Gross steel	2025	5.659 million tonnes
Acerinox High Performance Alloys melt	2025 output, VDM and Haynes once	2025	83 thousand tonnes
Acerinox High Performance Alloys finish	2025 output, all forms, worldwide	2025	47 thousand tonnes
Outokumpu stainless deliveries	Stainless steel	2025	1,751 thousand tonnes
Avesta high nickel programme	Specialty stainless pocket, timed below	2026 plan	about 13 thousand tonnes
Aperam Alloys and Specialties	Different book from High Performance Alloys	2025	61 thousand tonnes
EU flat apparent consumption	Mixed PRODCOM 24452200, upper bound	2024	57.8 million kilograms
German flat apparent consumption	Same mixed code	2024	30.4 million kilograms
EU bar apparent consumption	PRODCOM 24452100, rounded	2024	5.4 million kilograms

Sources: Mine and primary rows (International Nickel Study Group, 2026a, 2026b; U.S. Geological Survey, 2026s), stainless melt (worldstainless, 2026a, 2026b), Acerinox output and sales (Acerinox, 2026c, 2026d, 2026e), Outokumpu (Outokumpu Oyj, 2026a, 2026c), Aperam (Aperam S.A., 2026), apparent consumption (Eurostat, 2025b, 2025c, 2025d). Chapters 7, 8 and 9 use this table and do not restate the ladder.

Apparent consumption of European nickel mill forms

Where Eurostat publishes sold production, imports and exports for the same code, apparent consumption equals sold production plus imports minus exports (Team, 2026d). For EU nickel plate, sheet, strip and foil (PRODCOM 24452200) in 2024, sold production was 80.9 million kilograms, imports were 14.4 million kilograms and exports were 37.5 million kilograms, so apparent consumption was 57.8 million kilograms (Eurostat, 2025d). The sold production leg carries Eurostat's estimated flag, EU trade in this dataset is extra EU trade, and hot band sold to an EU reroller and then sold again as cold strip can sit inside sold production twice. The code mixes pure nickel and nickel alloy and does not report contained nickel, and the four digit customs split between unalloyed sheet and nickel alloy sheet is not yet in the source log, so 57.8 million kilograms is an upper bound on a mixed code.

The value columns show why that upper bound is not a specialty price. EU sold production was about EUR 14,476 per tonne, German sold production was about EUR 8,788 per tonne, and EU imports were about EUR 47,824 per tonne, all team estimates from the logged value and quantity cells (Team, 2026l, 2026m, 2026n). Acerinox High Performance Alloys sales of EUR 1,683 million on 47 thousand finished tonnes are about EUR 35,809 per finished tonne, which is a different book from the mixed flat code (Acerinox, 2026e; Team, 2026i). German sold production jumped from 15,870 tonnes in 2019 to 36,477 tonnes in 2020 while the logged value stayed near EUR 177 million to EUR 191 million, so the quantity series has a break that a buyer should not read as a doubling of alloy demand (Team, 2026k).

Long products have the opposite trade position, so flat and bar are planned as two balances. EU apparent consumption of nickel bar, rod and wire was 5,380,347 kilograms in 2024, from sold production of 2.5 million kilograms and imports of about 5.6 million kilograms (Eurostat, 2025b). The sold production cell is rounded on a 500,000 kilogram base, so apparent consumption can sit at about 5.1 million to 5.6 million kilograms, and the net importer conclusion holds inside that band (Team, 2026f). Exports of about 2.70 million kilograms sit above the rounded production figure, which means stock or an estimate that is not a census. Special Metals at Hereford is outside the EU27 customs territory, and HS 7502 unwrought nickel, where rotating billet is often declared, is outside this mill product identity (United Nations Statistics Division, 2026a).

Tube has three logged legs and still does not support a capacity recommendation. EU sold production of nickel tubes, pipes and fittings was 21.0 million kilograms in 2024, imports were 4.7 million kilograms and exports were 16.5 million kilograms, so apparent consumption was 9.2 million kilograms, and the sold production cell is a round estimate (Eurostat, 2025a). Subsea tube and chemical tube stay options with no tonnes until a plant route and a named grade are the company's own. The unit values and the bar band are set beside the tonnes in

Chapter 6, which is why the next subsection only places output, trade and stainless melt on the same page.

Output, trade and stainless melt

The German export leg of the flat balance agrees with the customs record closely enough to give the two sources different jobs. Comtrade reports German exports of HS 7506 at 17,775 tonnes in 2024, and PRODCOM reports 18,087,152 kilograms for the export leg of code 24452200 (Eurostat, 2026a; United Nations, 2026). PRODCOM remains the base for the European balance and Comtrade remains the base for partner corridors, and neither series is qualified alloy demand.

Acerinox High Performance Alloys melted 83 thousand tonnes and finished 47 thousand tonnes in 2025, counting VDM Metals and Haynes once, and those figures are output in a weak chemical and oil and gas year rather than nameplate (Acerinox, 2026d, 2026g). Finished product is worldwide and already includes the United States plants in the same division, so it is not a European ceiling and it is not compared with EU flat apparent consumption until an alloy split exists. A vacuum induction heat and a later remelt of the same metal would double count if the 83 thousand tonnes were added to stainless melt, which is why Appendix A4 asks for the filing footnote and for nameplate by plant and by route. The furnace and finishing steps aimed at 2028 are capacity that is not inside the 2025 tonnes (Acerinox, 2026f).

Outokumpu's about 13 thousand tonnes is a specialty stainless pocket at Avesta, inside 2025 deliveries of 1,751 thousand tonnes, and it is not 13 thousand tonnes of Alloy 625 or Alloy C-276 (Outokumpu Oyj, 2026a, 2026c). Remelt equipment is planned for early 2028, and greenfield melt and casting follow a decision in the first half of 2027 with production at the end of 2029 (Outokumpu Oyj, 2026b). Duplex, 904L and 254 SMO are stainless grades, and the overlap with a nickel alloy mill is mainly on nickel predominant grades such as Alloy 825, which is evidence of competitive intent at the boundary rather than current chapter 75 plate. Aperam Alloys and Specialties shipped 61 thousand tonnes in 2025 at a printed average selling price of EUR 17,475 per tonne, which is a different book and stops a reader treating Acerinox as the only European specialty mill (Aperam S.A., 2026). Chromium feeds the stainless system and is the next profile for that reason.

4.2 Cobalt

Cobalt shares aerospace, energy and medical applications with nickel alloys, and it does not share the European mill balance just calculated for nickel plate and bar. World mine production of contained cobalt was an estimated 310,000 tonnes in 2025, of which the Democratic Republic of the Congo accounted for about 230,000 tonnes, and refined production was a separate 240 thousand tonnes with China the leading refined producer

(Cobalt Institute, 2026a; U.S. Geological Survey, 2026h, 2026i). The price that matches the metal stage is the estimated London Metal Exchange cash average of USD 15 per pound in 2025, which is not a hydroxide price and not an alloy selling price (U.S. Geological Survey, 2026j). The commercial form a mill can buy for superalloys is alloy grade metal, and superalloys took 20 thousand tonnes of contained cobalt in 2025, while aviation use was 11,499 tonnes, of which 72 percent was commercial (Cobalt Institute, 2026b, 2026c).

Availability through 2027 is a quota rather than a mine capacity number. Export quotas replace the 2025 ban at up to 96,600 tonnes of contained cobalt a year in 2026 and 2027, including a strategic reserve (U.S. Geological Survey, 2026k). No European mill apparent consumption of cobalt is stated, because the PRODCOM code previously mapped to cobalt is labelled bismuth and cadmium and has empty value arrays (Eurostat, 2026d). Section 5.4 therefore uses the 11,499 tonnes only as a cobalt indicator, and a path of about 16,100 tonnes in 2035 is a team illustration that holds 2025 intensity constant with the fleet interpolation, not a Cobalt Institute forecast and not the nickel alloy market (Team, 2026a). German HS 8105 trade in 2024 was a small net export of a mixed matte, unwrought, powder and article code, which is the corridor Chapter 6 reports and not a superalloy balance (United Nations Statistics Division, 2026e). Chromium is next, because it is the bulk alloying element in the stainless melt that Table 2 keeps apart from this metal.

4.3 Chromium

Chromium enters the same plants as nickel, but the traded forms are chromite ore, ferrochromium and chromium metal, and a tonne of any one of them is not a tonne of stainless steel or of nickel plate. World chromite mine output is estimated at 51 million tonnes gross weight in 2025, and the demand pool that absorbs most ferrochromium is the stainless melt already shown in Table 2 (U.S. Geological Survey, 2026e; worldstainless, 2026a). The United States ferrochromium price, on a chromium content basis, is estimated at USD 1.60 per pound in 2025, while chromium metal is estimated at USD 5.90 per pound, so the two quotations are not one price (U.S. Geological Survey, 2026f, 2026g).

The scale check against nickel flat product stays the one in Table 2. EU stainless melt of 5.659 million tonnes of steel against EU nickel flat apparent consumption of about 58 thousand tonnes of a mixed code is a gross tonne comparison only, and it does not convert melt into contained chromium and does not set a high performance market (Eurostat, 2025d; worldstainless, 2026b). Outokumpu's stainless deliveries and the timed Avesta pocket are already in that table, and they stay in stainless rather than being retold here as nickel alloy plate (Outokumpu Oyj, 2026a). Molybdenum, the next alloying metal, raises the same distinction between a mine balance and a mill product.

4.4 Molybdenum

Molybdenum follows chromium as an alloying addition whose mine stage is a different unit from the nickel mill products in section 4.1. World mine output of contained molybdenum was an estimated 260,000 tonnes in 2025 (U.S. Geological Survey, 2026o). Part of that output comes from primary molybdenum mines and part is recovered with copper, so the two routes have to be read separately before any availability claim is made. United States apparent consumption of molybdenum is estimated at 34,000 tonnes of contained metal in 2025, built from production, trade and stocks (U.S. Geological Survey, 2026p). That national metal balance is a different identity from the European mill product apparent consumption used for nickel plate and bar.

The price that matches this metal is the United States molybdic oxide benchmark, estimated at 51 dollars per kilogram in 2025, up from 47.72 dollars in 2024 (U.S. Geological Survey, 2026q, 2026r). That quotation is for oxide, so ferromolybdenum or a molybdenum bearing alloy needs its own price before anyone treats the oxide figure as a selling price. The commercial form inside a corrosion grade is that alloy addition, not the oxide benchmark, which is why titanium is taken next as sponge and approved melt rather than as another nickel flat identity.

4.5 Titanium

Titanium is the metal that cannot stay a caveat beside an aerospace priority, because the same engine makers buy titanium mill product and the constraint is sponge and approved melt rather than the nickel flat balance. World sponge production is estimated at 370,000 tonnes in 2025, of which China produced an estimated 260,000 tonnes, and the United States has no operating sponge capacity in that year (U.S. Geological Survey, 2026w, 2026x, 2026y). The price at the sponge stage is the estimated United States landed duty paid import unit value of USD 12 per kilogram in 2025 (U.S. Geological Survey, 2026z). The commercial form a mill ships is wrought bar, sheet or tube after melting, and EU sold production of wrought titanium was 22,617 tonnes in 2024 against unwrought sold production of only 1,110 tonnes, which is a mixed code and not a qualified aerospace balance (Eurostat, 2024a, 2024b).

No European mill apparent consumption is asserted as aerospace demand, because those PRODCOM codes still mix forms. German HS 8108 trade in 2024 was a net import, led by value from the Russian Federation and the United States, and that corridor is reported in Chapter 6 without being converted into a Münvest tonne (United Nations Statistics Division, 2026f). Japan and China are the sponge volume centres, and a qualified melt route remains the gate for engine billet, which is why tungsten is taken next as a separate carbide chain rather than as another reading of this sponge balance.

4.6 Tungsten

Tungsten's largest use is cemented carbide, so a mine statistic does not describe an alloy shipment and a carbide price does not describe a nickel mill product. World mine production was an estimated 85,000 tonnes of contained tungsten in 2025, of which China accounted for an estimated 67,000 tonnes, and United States mine production was zero (U.S. Geological Survey, 2026aa). The price that names its stage is the estimated Rotterdam concentrate average of USD 380 per dry metric ton unit of tungsten trioxide in 2025, after a path during the year from USD 266 to USD 551 for concentrate and from USD 331 to USD 675 for ammonium paratungstate (U.S. Geological Survey, 2026aa). About 60 percent of United States consumption goes to cemented carbide parts, which is the commercial form that dominates volume, while steels and superalloys are a smaller first use (U.S. Geological Survey, 2026aa).

China required an export licence from 4 February 2025 on specified tungsten items, including ammonium paratungstate, and that control does not state a quota in tonnes (Ministry of Commerce, 2025). EU sold production of tungsten powders and unwrought metal is logged as a rounded cell and is not used here as alloy demand (Eurostat, 2026c). No European mill apparent consumption of a tungsten alloy form is stated. Sandvik's powder business is evidence that recycling and powder sit with the carbide chain, not that a nickel alloy mill should add tungsten capacity from the mine total (Sandvik, 2026). The remaining metals are grouped next, with the same refusal to add unlike stages together.

4.7 Other Selected Metals

4.7.1 Zirconium

Zirconium shows again why mineral output and specialty metal supply stay apart. World mine production of zirconium mineral concentrates is estimated at 1,200 thousand tonnes gross weight in 2025, led by Australia, and the leading end use of zircon is ceramics rather than implant bar (U.S. Geological Survey, 2026ab, 2026b, 2026c). The price at the mineral stage is an estimated premium grade of USD 1,800 per tonne, cost insurance and freight China, in 2025 (U.S. Geological Survey, 2026ac). EU sold production of unwrought zirconium and powders was 4,000 kilograms in 2024 with a rounded flag, so no European mill apparent consumption is stated as implant demand (Eurostat, 2026b). Oxidized zirconium joint parts are Smith and Nephew's OXINIUM route, not a generic zirconium bar inferred from zircon sand, and the VDM portfolio reference remains an application benchmark only.

4.7.2 Copper

Copper keeps mine production, refining and a cathode price on separate lines, because none of them is a high performance plate order. World mine production of recoverable copper was a

rounded 23 million tonnes in 2024, and world refinery production is estimated at a rounded 29 million tonnes in 2025 (U.S. Geological Survey, 2026l, 2026m). The 2025 mean of monthly Pink Sheet prices was USD 9,947 per tonne, which is a refined metal benchmark and not a conversion price for alloy plate (World Bank, 2026b). Copper and aluminium plate workings stay out of the recommendation until the same three leg standard used for nickel is the basis of a capacity case, because a commercial assessment still needs an identified product, customer and route.

4.7.3 Aluminium

Aluminium is a comparison point for transport and electrical uses, and primary metal is not a proxy for nickel or cobalt alloys. World primary smelter production is a rounded 74 million tonnes in 2025, with China at 45 million tonnes, and the Pink Sheet annual average was about USD 2,632 per tonne (U.S. Geological Survey, 2026d; World Bank, 2026a). Bauxite, alumina, primary metal and recycled metal stay separate stages, and the International Aluminium Institute's 73.8 million tonnes for 2025 sits close to the rounded USGS total without changing that separation (International Aluminium Institute, 2026). Aluminium demand is not used as a substitute for the nickel balances in Table 2.

4.7.4 Manganese

Manganese is an alloying addition in steel and in some nickel grades, and it is also a battery chemical, so ore tonnes are not a mill product franchise. World mine production on a manganese content basis was a rounded 20 million tonnes in 2025, led by South Africa and Gabon, and the United States ore price was about USD 4.5 per metric ton unit of manganese content (U.S. Geological Survey, 2026n). The International Manganese Institute reports 21.4 million tonnes metal contained for the same year, which is a second mine series and not a ferroalloy shipment (International Manganese Institute, 2026). No European apparent consumption of a manganese mill form is used in the recommendation. For Münvest, the task is to establish which grades overlap with the intended book before any conclusion is drawn from aggregate demand.

4.7.5 Additional Metals

Other metals may be included where a specific application or a sourcing dependency makes them relevant, and this category stays an extension rather than a residual total of unlike materials. Iron and steel provide context for several of the metals above, and finished steel and stainless steel are not added to the output of their alloying constituents. The reason is Table 2: specialty nickel flat apparent consumption is an upper bound in the tens of thousands of tonnes, and stainless melt is tens of millions of tonnes.

Company comparisons across the profiles need the same separation of unlike businesses. On grades such as 625, 718, Alloy C-276, 188 and 230, Acerinox High Performance Alloys is one line, with VDM in Germany and Haynes in the United States inside that line, and their tonnes are not added again to the 83 thousand and 47 thousand already counted (Acerinox, 2026d). ATI, Carpenter Technology, Special Metals, Alleima and Aperam Alloys are relevant to particular forms, and their total revenues do not establish comparable market shares. Chapter 5 now asks which industries buy the forms these profiles have kept distinct.

5. Industry Needs

The seven industry blocks that follow keep the application evidence and examine demand from the industry driver through the component, the material, the delivery form and the buyer. Their role is to show how the metals in Chapter 4, including the basket in Table 2, become orders a buyer can place. Industry revenue measures economic activity in the sector, and it is not read here as addressable alloy demand, just as a named account stays a prospect until a relationship is documented. To limit overlap in the scenario model, electrolysers and fuel cells are assigned to energy, process vessels to chemicals, and sensors and magnetic components to electronics.

Each sector closes with the same four verbs that Chapter 8 uses, so the chapters do not issue seven different mandates. The company should qualify for repeating orders in aerospace billet, approved sheet and repair wire, and in chemical replacement plate and welding products where a named grade clears the capacity rule. The company should keep and harvest oil and gas accounts only where an approval the company itself holds supports the margin, and automotive accounts that are already profitable. The company should treat electrolysers, powders and medical implant bar as options against a financed programme or a regulatory path that is the company's own. The company should stop adding effort on undifferentiated stainless, on China implant tenders, and on any electrolyser or powder programme without a financed buyer. Chapter 7 then turns those readings into demand sensitivities, and it does not multiply them by an unmeasured share of the mixed flat code.

5.1 Automotive

Automotive is a harvest market for engine and exhaust parts, because the number of cars without a plug falls in the team estimate while a car with a plug can still carry an engine. Table 3 takes world car registrations from 77.6 million in 2025 to 85.7 million in 2035 and cars without a plug from 58.2 million to 42.9 million, which is a loss of about a quarter of that set and not the loss of every engine part (European Automobile Manufacturers' Association, 2026; Team, 2026e). Magnetic sensors and magnetic alloys are not counted again here, because section 5.0 assigned them to electronics so the scenario model would not count them twice.

The applications below are therefore the engine and exhaust book, and the close keeps only the accounts that are already profitable.

5.1.1 Historical and Current Market Overview

In 2025 the world built 96.4 million motor vehicles, China built 34.53 million of them, and passenger car registrations were 77.6 million (European Automobile Manufacturers' Association, 2026; International Organization of Motor Vehicle Manufacturers, 2026a, 2026b). Those are activity totals for the vehicle industry, and they are not tonnes of nickel alloy, so they sit beside the basket in Table 2 only as the driver of the component set.

The 2035 car figures in Table 3 are a team estimate that holds the electric share near half of new cars and lets registrations grow by 1 percent a year, and the October 2026 committee vote on the European rule is left out of that estimate because it is not a logged value row. A hybrid can still need a turbocharger, a valve or an exhaust part, which is why the upside in Chapter 7 is no longer a flat zero (Team, 2026e). The table is the physical base for that rate, and the applications that follow say which of those parts are worth harvesting.

Table 3: Car sales, 2025 and 2035

Measure	2025	2035	Change
Cars sold (million)	77.6	85.7	+10 percent
Share electric	about 25 percent	about 50 percent	team estimate
Cars without a plug (million)	58.2	42.9	-26 percent

Sources: Registrations in 2025 are taken from ACEA (European Automobile Manufacturers' Association, 2026), and the 2035 column together with the cars without a plug is a team estimate that applies the shares in this table to a 1 percent a year path rather than an official forecast (Team, 2026e).

5.1.2 Key Applications and Opportunities for the Company

The benchmark application, from the VDM Metals portfolio, covers nickel alloys, stainless and catalyst foil in turbochargers, valves, spark plugs, glow plugs, exhaust parts and airbag components. What Münvest would need before this is an order is its own grade, form, approval and buyer, and shipment, approval and share are unconfirmed. Eight of the eleven catalogue lines sit on the engine and exhaust, and that count is a reading of the benchmark rather than a count of Münvest approvals.

Table 4: Benchmark automotive applications, outlook to 2035

Outlook	Applications	Materials
Points to section 5.6	Sensors and magnetic shielding	Nickel iron alloys
Stable if already profitable	Airbag parts and catalyst foil	Nickel alloys and iron chromium aluminium foil
Harvest, not a growth list	Turbochargers, valves, spark plugs and exhaust parts	Nickel alloys and heat resistant stainless
Stop	Glow plugs and the diesel only set	Nickel alloys

Sources: The application list is the VDM Metals portfolio used as a benchmark only. The outlook column is the team estimate that matches Chapter 8, and it is not a VDM shipment.

Garrett Motion reported net sales of USD 3,584 million in 2025, with gasoline at USD 1,592 million and diesel at USD 837 million, and BorgWarner reported Turbos and Thermal sales of USD 5,772 million (BorgWarner Inc., 2026; Garrett Motion Inc., 2026). Those figures name the buyers and the year, and they do not name the alloy mill or the tonnes. The contest in European spark plug and sensor strip is Aperam Alloys, Proterial, Nippon Yakin and Acerinox High Performance Alloys, counted once, while Carpenter and Special Metals are minor in that European strip (Acerinox, 2026d; Aperam S.A., 2026).

What the company should do: the company should keep and harvest automotive accounts that are already profitable, including turbocharger and exhaust business while cars without a plug are still built, and it should take glow plugs and the rest of the diesel only set off the growth list now. Magnetic alloys for sensors are pursued in section 5.6 rather than here, because that assignment stops the scenario model from counting the same component twice.

5.2 Oil and Gas

Oil and gas is a keep market, because upstream spending in 2025 is a replacement flow and the alloy case depends on an approval the company itself holds. World Energy Investment 2025 puts combined upstream oil and gas at just under USD 570 billion, and about 40 percent of that combined total slows decline at existing fields (International Energy Agency, 2025g, 2025h). That 2025 level is not a printed 2035 total, and it is not the wider 2026 supply boundary used in the next paragraph. The applications that follow are therefore a harvest of sour service work, not a proposal to add tube or bar capacity.

5.2.1 Historical and Current Market Overview

The sentence that gas spending is at a ten year high and oil spending has fallen for three years belongs to 2026 supply investment on the wider boundary, where natural gas supply investment is USD 330 billion and oil supply investment is below USD 500 billion, including infrastructure (International Energy Agency, 2026k). Those figures are not attached to the just under USD 570 billion upstream total. World Energy Outlook 2025 does not print a path that stays at USD 570 billion or rises to about USD 670 billion, and Current Policies is about USD 65 billion a year above Stated Policies in the detailed chapter (International Energy Agency, 2025i).

Table 5: Oil and gas activity, 2025 and the logged outlook

Measure	2025 reading	Later reading
Upstream oil and gas investment	Just under USD 570 billion	No printed 2035 total. Current Policies is about USD 65 billion a year above Stated Policies
Oil consumption	103 million barrels a day	Stated Policies peaks around 2030
LNG	560 billion cubic metres of demand in 2024	Current Policies trade of 880 billion cubic metres in 2035

Sources: Upstream investment (International Energy Agency, 2025g, 2025h, 2025i). Oil consumption is the Energy Institute series for 2025 and the peak is the World Energy Outlook Stated Policies case (Energy Institute, 2026; International Energy Agency, 2025j). LNG demand and Current Policies trade are (International Energy Agency, 2025k). None of these cells is alloy demand.

5.2.2 Key Applications and Opportunities for the Company

Regional investment in 2026 is uneven on the same upstream boundary, and it narrows where a keep decision is even worth a visit. Sub Saharan Africa is nearly USD 24 billion and Central and South America nearly USD 68 billion, while Middle East upstream spending is down by around 1 percent (International Energy Agency, 2026l). That geography can guide which accounts to keep, and it is not a reason to add melting capacity. The benchmark applications below say which forms are even eligible for that keep decision.

The benchmark application, from the VDM Metals portfolio, covers nickel alloy bar, plate, billet, welding wire and strip for sour service tools, flanges and lined pipe. What Münvest would need before this is an order is a sour service approval in its own name, because a NACE MR0175 or ISO 15156 listing names the producer and the route and does not travel with a

catalogue. Shipment, approval and share are unconfirmed, and no approval count is taken from the benchmark.

Table 6: Benchmark oil and gas applications, outlook to 2035

Outlook	Applications	Product forms
Keep, if the company holds the approval	Well tools, flanges and casing	Bar, plate and billet
Option, no tonnes	Lined pipe and subsea tube	Wire, strip and tube
Stop	Any tube capacity case	Tube

Sources: The application list is the VDM Metals portfolio used as a benchmark only. Tube is off the pursue list because section 4.1 logs the three PRODCOM legs and still withholds a capacity recommendation (Eurostat, 2025a).

The buyers named in the public record include SLB, TechnipFMC and BUTTING, and BUTTING reported turnover of EUR 880 million in 2025, which is a fabricator total and not an alloy tonne (BUTTING Group, 2026; SLB, 2026). The contest on 625 and similar grades is Acerinox High Performance Alloys as one line, together with Special Metals, Alleima, Nippon Steel and Tubacex, and the 2015 combined name is not used (Acerinox, 2026d). Thinner linings reduce metal per project, which is another reason the keep decision is about margin on an existing approval and not about more tonnes.

What the company should do: the company should keep oil and gas accounts where a sour service approval the company itself holds supports the margin, and it should not add tube capacity or treat bar as a new investment case. Welding wire for lined pipe can stay inside that keep decision only as a product the mill already makes, and the commercial test of bar is the one set in Chapter 8 against the logged European position.

5.3 Energy and Environmental Technology

Energy and environmental technology is an option market, because electricity growth and announced hydrogen are activity totals and the alloy order exists only where a programme is financed. Global electricity demand grew 4.4 percent in 2024 and 3 percent in 2025, and the Electricity 2026 forecast is 3.6 percent a year over 2026 to 2030 (International Energy Agency, 2026e). Those rates are not applied to nickel alloy tonnes, so the component paths below are qualified one at a time and unfunded hydrogen stays on the stop list.

5.3.1 Historical and Current Market Overview

Hydrogen shows the gap between an announcement and a bankable order, because global demand exceeded 100 million tonnes in 2025, low emissions output was about 1 million tonnes, the announced 2030 pipeline is 27 million tonnes, and installed electrolysis exceeded 4 gigawatts (International Energy Agency, 2026f, 2026g, 2026h, 2026i). Committed low emissions output, meaning projects operating, under construction or at a final investment decision, is 4.3 million tonnes a year by 2030 (International Energy Agency, 2026j). Output above 6 million tonnes is the review's conditional case if further projects reach that decision, and it is not a stock already committed. Chapter 7 uses the same split, which is why the table below does not treat the pipeline as alloy demand.

Table 7: Energy activity and the forms that could become an order

Reading	Logged figure	What it is not
Electricity demand growth	4.4 percent in 2024, 3 percent in 2025, 3.6 percent a year from 2026 to 2030	Not alloy tonnes
Committed low emissions hydrogen	4.3 million tonnes a year by 2030	Not the 27 million tonne pipeline
Conditional case	Just over 6 million tonnes if further projects are decided	Not capacity to build against
Alkaline buyer, Germany	thyssenkrupp nucera green hydrogen orders EUR 26 million in 2024/25	Not a strip contract

Sources: Electricity growth is taken from the same outlook for all three rates (International Energy Agency, 2026e), hydrogen figures are (International Energy Agency, 2026f, 2026g, 2026h, 2026i, 2026j), and Nucera orders are (thyssenkrupp nucera, 2025b).

5.3.2 Key Applications and Opportunities for the Company

The benchmark application, from the VDM Metals portfolio, names Nickel 201 strip for alkaline electrolyser cathodes and Crofer 22 for solid oxide interconnects. What Münvest would need before this is an order is a financed buyer, a form the mill already finishes, and a contribution that survives the Chapter 7 downside. The account is unconfirmed wherever the public record names the alloy and not the purchase order.

thyssenkrupp nucera, in Germany, is the named alkaline buyer in the company set, and its green hydrogen orders fell to EUR 26 million in fiscal 2024/25 from a much higher prior year while chlor alkali orders were EUR 322 million (thyssenkrupp nucera, 2025a, 2025b). The annual report names nickel and titanium as sourced minerals and does not give tonnes, and the company abandoned solid oxide stack mass production in 2026 while keeping a pilot (thyssenkrupp nucera, 2026). Siemens Energy told investors in November 2025 that gas

turbine capacity is sold out until fiscal 2028, with a new unit backlog of about 78 gigawatts, and the metal form there is superalloy casting and forging rather than Nickel 201 strip (Siemens Energy, 2025). Bloom Energy, in the United States, reported fiscal 2025 revenue of about USD 2.0 billion on energy servers associated with ferritic stainless, and any nickel strip account remains unconfirmed (Bloom Energy, 2026). Acerinox High Performance Alloys is one competitor on the turbine grades, through Haynes, and it is the same group already counted at 47 thousand finished tonnes (Acerinox, 2026d, 2026g).

What the company should do: the company should treat electrolyser strip and powders as an option against a financed programme, and it should stop adding effort where the buyer is not financed. Industrial gas turbine hardware belongs with the aerospace qualification in section 5.4, because 718, 625, X and 230 are being pulled there, and it is not a second capacity case sized to electricity growth.

5.4 Aerospace

Aerospace is the first qualification priority, because new engine hardware and shop visits are both pulling 718, 625, X and 230, and the mill sees billet, approved sheet and weld wire rather than the engine maker's margin. The passenger fleet path of about 3.4 percent a year, from 23,310 aircraft at the end of 2025 to about 32,600 in 2035, is a team interpolation between the Airbus 2025 and 2045 stocks and is not an Airbus 2035 figure and not an alloy tonne forecast (Airbus, 2026e; Team, 2026j). The published Airbus figure for 2035 that this chapter uses is the 82 percent share of new generation aircraft (Airbus, 2026f). Cobalt in aviation is an indicator in section 4.2, and it is not resized here into a nickel market.

5.4.1 Historical and Current Market Overview

Airbus and Boeing delivered 793 and 600 commercial aircraft in 2025 (Airbus, 2026d). Over 2026 to 2045 Airbus expects 42,060 new passenger aircraft of at least 100 seats, while Boeing expects 43,625 deliveries that include 1,435 regional jets and 930 freighters, so the two totals are not the same perimeter (Airbus, 2026g; Boeing, 2026c). Engine output has to be read on the same careful basis, because Safran delivered 1,802 LEAP engines in 2025, up 28 percent from 1,407, and full year 2025 commercial engine deliveries at GE Aerospace were 2,386, which already contains the LEAP engines and must not be added to them (GE Aerospace, 2026a; Safran, 2026a). The commercial engine increase of 25 percent, 1,989 against 1,592, is the first nine months of 2026, and defence engines of 635 against 490 are the 2025 increase (GE Aerospace, 2026b, 2026e).

Table 8: Aerospace activity, with the 2035 fleet as a team interpolation

Measure	Today	Later reading
Passenger aircraft of at least 100 seats	23,310 at the end of 2025	About 32,600 in 2035, team interpolation, and 82 percent new generation
Cobalt used in aviation	11,499 tonnes in 2025, a cobalt indicator	About 16,100 tonnes if intensity is held constant, team estimate, not a nickel market
LEAP engines	1,802 in 2025, up 28 percent from 1,407	Not added to the 2,386 commercial engines delivered in 2025
GE commercial engines	2,386 in full year 2025	1,989 in the first nine months of 2026, up 25 percent on 1,592
Aviation services	USD 209 billion in 2026	USD 422 billion in 2045

Sources: Fleet and the new generation share are (Airbus, 2026e, 2026f; Team, 2026j), cobalt is (Cobalt Institute, 2026c; Team, 2026a), engines are (GE Aerospace, 2026a, 2026b, 2026e; Safran, 2026a), and services are (Airbus, 2026h, 2026i).

Shop visits are the larger profit pool at the engine makers, and they are a different purchase from the original billet. GE Commercial Engines and Services earned a 26.6 percent margin on USD 33,314 million, with services at 75 percent of that segment, and Safran propulsion earned 23.0 percent on EUR 15,668 million, with services up 30 percent in dollar terms and civil spare parts up 17.6 percent in dollar terms (GE Aerospace, 2026c; Safran, 2026b). MTU's commercial maintenance revenue was EUR 6.0 billion against OEM revenue of EUR 2.9 billion in 2025 (MTU Aero Engines, 2026). Airbus puts the services market at USD 209 billion in 2026 and USD 422 billion in 2045, and it describes off wing work by growth of 3.7 percent a year and by a materials share above 85 percent (Airbus, 2026h, 2026i). HS 8411 includes parts, so the 2024 export ranking is not a ranking of complete engines only. The United Kingdom exported USD 39.8 billion, France USD 22.0 billion and the United States USD 14.1 billion (United Nations Statistics Division, 2026i). The route in the next subsection is what a wrought mill can offer into that flow.

5.4.2 Key Applications and Opportunities for the Company

Hot gas path blades are cast, single crystal or directionally solidified, and the casters are Howmet and PCC, so a wrought mill does not sell blade bar into LEAP. Discs and rings are billet and then a forging, and in Europe that forger is Aubert and Duval, while qualified melters already on rotating grades include ATI, Carpenter, Special Metals and Aubert and Duval. Combustor and casing sheet is the Haynes book of 188, 230, 625, X and similar grades,

and Haynes is Acerinox High Performance Alloys, so the competitor and the benchmark portfolio are the same group (Acerinox, 2026d, 2026g). The offer is billet for forged discs through a named forger, sheet only where this mill is already approved, and weld wire for shop visits. Powders and the laboratory alloy GRX-810 are a watch item, because that alloy has its own atomisation route and is not a powder volume for 2035.

Table 9: What a wrought mill can offer into aerospace

Outlook	Applications	Form
Qualify	Forged discs and rings, through a named forger	Billet
Qualify, only if already approved	Combustor and casing	Sheet
Qualify	Shop visits	Weld wire
Watch	Printed parts and the hottest static hardware	Powder, and ceramic matrix composites on static parts only

Sources: The route is the team reading of the engine supply chain in this section. Services growth and the materials share are (Airbus, 2026i). Ceramic matrix composites are limited to the hottest static parts and do not retire the disc.

The buyers are GE Aerospace, Safran, Rolls-Royce, MTU and Pratt and Whitney (GE Aerospace, 2026c; MTU Aero Engines, 2026; Rolls-Royce, 2026; Safran, 2026b). Approval for a rotating grade is an AMS specification, a NADCAP approval and an engine test, and it is measured in years. A shortage in 2025 is a weld wire and second source conversation for a grade the mill already makes, and it is not an entry ticket to a LEAP disc (GE Aerospace, 2026d). ATI takes about 68 percent of sales from aerospace and defence, and Carpenter about 65 percent, while Acerinox High Performance Alloys finished 47 thousand tonnes in 2025 (Acerinox, 2026d; ATI Inc., 2026a; Carpenter Technology, 2026c). Carpenter specialty alloys at 27.7 percent and ATI high performance materials at 23.6 percent are cycle margins, and ATI advanced alloys at 16.3 percent is the closer comparison for a new source (ATI Inc., 2026b; Carpenter Technology, 2026d). The one competitor line is Acerinox High Performance Alloys, together with Aubert and Duval, Special Metals, ATI, Carpenter, Howmet and PCC for castings, Proterial, and Daido on shafts. Chinese mills matter less for a vacuum remelted disc billet than they do for standard 625 plate.

What the company should do: the company should qualify for repeating orders in billet sold through a named forger, in sheet only where the mill is already approved, and in repair wire, and it should treat a 2025 shortage as a second source conversation rather than as an entry to a disc. Powders stay an option until a printer programme is financed, and ceramic matrix composites are tracked only as a substitute for the hottest static parts. Industrial gas turbines

use the same grades and sit on this qualification list, with Haynes the part of Acerinox High Performance Alloys that faces that demand and VDM the part that faces the weak chemical and oil and gas year (Acerinox, 2026g).

5.5 Chemical Processing

Chemical processing is a qualification market for replacement plate and welding products, and growth in chemical sales does not create an equipment order. Cefic reports EU27 chemical industry turnover of EUR 635 billion in 2024, with the EU share of world sales at 13 percent, down from 21 percent in 2009, and those figures describe chemicals rather than alloy tonnes (European Chemical Industry Council, 2025a, 2025b). Acerinox described chemical processing as significantly weaker in 2025, which is why new build is not the base case and replacement of a named grade is (Acerinox, 2026h). The table below keeps the grade list that makes this the most specific sector text, and it adds the buyer test the other sectors now use.

5.5.1 Historical and Current Market Overview

European weakness supports a cautious reading of new plant, while the installed base still needs maintenance when a grade fails in a stated medium. Regional sales shares can guide which fabricators to call, and they are not a regional alloy forecast. Tube is not a capacity recommendation, because the three PRODCOM legs in section 4.1 are a rounded mixed code and subsea and chemical tube stay options with no tonnes (Eurostat, 2025a). The applications that follow are therefore plate and weld products, read against alloyed flat once that customs split exists, and not against the 57.8 million kilogram ceiling.

Table 10: Chemical processing, benchmark grades and the order test

Grade or form	Duty in the benchmark	Order test
Alloy 59, Alloy C-276, Alloy 625 plate	Acid, caustic and heat exchanger service	Named grade, named medium, replacement or a specified supplier
Nickel 200 and 602 CA	Caustic and high temperature furnace service	Same test, separate from commodity 625
Welding consumables matching the plate	Fabricated equipment	Repeat quantity before a second sample
Zirconium solutions	Duties stainless does not meet	Benchmark only, not a Münvest qualification
Stainless or a lining	Duty already met	Stop

Sources: The grade list is the VDM Metals portfolio used as a benchmark only, the weaker 2025 chemical year is the Acerinox commentary (Acerinox, 2026h), turnover is (European

Chemical Industry Council, 2025b), and tube tonnes are withheld (Eurostat, 2025a).

5.5.2 Key Applications and Opportunities for the Company

The benchmark application, from the VDM Metals portfolio, covers Nickel 200, Alloy 59, Alloy C-276, 625, 602 CA and zirconium solutions for acid and caustic service, heat exchangers, vessels, pumps, valves and furnaces. What Münvest would need before this is an order is the actual medium, concentration, temperature and welding procedure, plus a fabricator trial that can close inside a year. Shipment, approval and share remain unconfirmed until the company holds the account. The direct route is usually the fabricator, while the plant operator influences the specification.

BUTTING, in Germany, reported turnover of EUR 880 million in 2025 on clad pipe, which shows a fabricator that already buys strip and wire for linings, and the alloy account for Münvest is unconfirmed (BUTTING Group, 2026). BASF reported group sales of EUR 59.7 billion in 2025, and the public record names the chemical operator rather than a plate order, so Alloy 59 and Alloy C-276 remain the specification with the account unconfirmed (BASF, 2026). Alleima took 16 percent of 2025 revenue from chemical and petrochemical customers and is a tube competitor rather than a Münvest qualification, which is why tube stays off the capacity list (Alleima AB, 2026). Standard 625 plate is contested by Chinese mills as well as by Special Metals and ATI, while a patented Haynes grade is a different contest inside Acerinox High Performance Alloys, counted once (Acerinox, 2026d, 2026g). Where stainless or a lining already meets the duty, the company should leave that tonne alone.

What the company should do: the company should qualify corrosion resistant plate and matching welding products for replacement equipment where a named grade clears the capacity rule, and it should stop undifferentiated stainless and any tube capacity case. Export projects are qualified through fabricators that can show funding, and a premium grade is proposed only where the documented service life and the actual medium justify it.

5.6 Electronics and Electrical Engineering

Electronics and electrical engineering is a qualification market for precision strip and wire, and semiconductor revenue is an activity indicator rather than a multiplier for alloy volume. World Semiconductor Trade Statistics reports sales of USD 795.6 billion in 2025, up 26.2 percent, and the implied 2024 level of USD 630.5 billion is the 2025 figure divided by that growth rate, not a printed 2024 cell (Team, 2026c; World Semiconductor Trade Statistics, 2026). Most of the soft magnetic and expansion book is iron predominant and sits in chapter 72, so one nickel flat balance cannot size it. Sensors and magnetic components are counted here, and not again in automotive, because section 5.0 made that assignment.

5.6.1 Historical and Current Market Overview

The 2025 chip expansion was uneven across end markets in the same release, and those end use splits are left as printed context rather than as a new calculation (World Semiconductor Trade Statistics, 2026). Strong chip sales do not establish the metal content of a package, a sensor or a production tool. Electrification adds a second driver through current measurement and heating elements, and the volume is unit production times alloy content times the share on the relevant design. The table below keeps those designs apart from the chip total so that a sales surge is not read as alloy demand.

Table 11: Electrical alloys, the code they ship in, and the order test

Book	Form and code	Order test
Expansion alloys 29-18 and 42	Strip, iron predominant, chapter 72	Named stamper, sample, then a repeat order
Crofer 22 and Aluchrom	Strip, iron chromium, chapter 72	Not sized by the nickel flat balance
Konstantan and copper nickel resistance alloys	Wire, chapter 74	Separate price and buyer from Alloy C-276 plate
Nickel 201 electrolyser strip	Unalloyed nickel in HS 7506	Financed programme, option in section 5.3
Soft magnetic grades	Strip	Counted here, not in automotive

Sources: The grade names are the VDM Metals portfolio used as a benchmark only. Semiconductor sales and the implied prior year are (Team, 2026c; World Semiconductor Trade Statistics, 2026). The code map is the capacity rule in Chapter 8.

5.6.2 Key Applications and Opportunities for the Company

The benchmark application, from the VDM Metals portfolio, lists expansion alloys 29-18 and 42, soft magnetic grades, and resistance alloys including HT 70, Konstantan and Aluchrom, for glass and ceramic seals, shielding, toroidal cores, current measuring modules and heating elements. What Münvest would need before this is an order is a stamper or core maker that already pays for the controlled property, a form the mill finishes consistently, and a contribution margin after the sample. The account is unconfirmed where the public record names the specification and not the purchase order.

Infineon, in Germany, reported revenue of EUR 14.7 billion in the year to 30 September 2025, and a 2022 note names copper alloy and iron nickel leadframes from named suppliers without a grade, so the strip account is unconfirmed (Digitimes, 2022; Infineon Technologies, 2025). TSMC reported 2025 net revenue of about USD 121 billion and names fabs in the United

States, Japan and Germany without naming a nickel alloy purchase (Taiwan Semiconductor Manufacturing Company Limited, 2026a, 2026b). Applied Materials reported fiscal 2025 net revenue of USD 28.4 billion and states that some parts come from a limited supplier group, again without naming a metal, so expansion alloys 29-18 and 42 stay a specification rather than an order (Applied Materials, Inc., 2025a, 2025b). The contest on this book is Aperam Imphy, Proterial, Nippon Yakin and VDM inside Acerinox High Performance Alloys, counted once, and Aperam Alloys and Specialties is the separate 61 thousand tonne book already in Table 2 (Acerinox, 2026d; Aperam S.A., 2026). Europe is the practical place to start because engineering contact is closer, and Asian qualification is justified only where the return clears the sample cap in Chapter 8. Semiconductor revenue growth does not set the alloy volume, which is why finishing capacity waits for a repeat order.

What the company should do: the company should qualify expansion, soft magnetic and resistance strip and wire only with buyers that already pay for the controlled property, and it should stop treating chip sales as alloy demand. Dedicated finishing is added only after sample acceptance, a repeat order and a contribution per bottleneck hour, on the same gate as the other two priorities.

5.7 Medical Technology

Medical technology is an option, not a growth mandate, because procedure counts rise on a logged path and the alloy mill ceiling is far smaller than implant maker revenue. The constant reporting set of hip and knee procedures rose from 1,797,396 in 2013 to 2,501,707 in 2023, a compound rate of 3.36 percent a year, and members reporting both procedures in 2023 recorded 2,751,272 (OECD, 2025a). A 2035 count of about 3.72 million is that logged rate applied to the constant set for twelve years, and it is a team estimate of procedures rather than of bar (Team, 2026b). The device market path is an illustration from 2023 to 2035 at 5.3 percent a year, which is a different window from the Chapter 7 medical rate that starts in 2025, and Appendix A3 already requires that label.

5.7.1 Historical and Current Market Overview

Germany recorded 292,143 hip procedures and 223,133 knee procedures in 2023, and Lithuania's hip count almost doubled over 2013 to 2023, so the procedure growth is real and uneven (OECD, 2025a). No official source used here publishes the metal weight per implant, so the procedure count stays the indicator and is not converted into tonnes. The device market illustration is kept in the table only to show that a 5.3 percent value path from 2023 is not the 2 percent application rate from 2025 in Chapter 7.

Table 12: Medical activity, with the device path labelled as an illustration from 2023

Measure	Logged reading	Later reading
Device market illustration	2023 base	2035 at 5.3 percent a year from 2023, illustration only, not the Chapter 7 rate
Hip and knee procedures, members reporting both	2,751,272 in 2023	Not a 28 country total
Constant reporting set	1,797,396 in 2013 to 2,501,707 in 2023	About 3.72 million in 2035 if 3.36 percent a year continues, team estimate
Carpenter medical sales	USD 224 million excluding surcharge in fiscal 2026, down 24 percent	USD 278 million including surcharge, down 21 percent

Sources: Procedures are (OECD, 2025a; Team, 2026b), Carpenter medical sales are (Carpenter Technology, 2026a), and the device row is the illustration required by Appendix A3.

Prices are under pressure in China, where central purchasing cut the prices of artificial hips and knees by 82 percent, which is why China tenders are on the stop list (National Healthcare Security Administration, 2022). The alloy opportunity, if any, is a bar the company itself has qualified, and the next subsection says how far the public record is from that point.

5.7.2 Key Applications and Opportunities for the Company

The benchmark application, from the VDM Metals portfolio, covers cobalt chromium and zirconium bar for joints, stainless wire for instruments, nickel iron strip for imaging equipment, and cobalt chromium powder for printed implants. What Münvest would need before this is an order is an implant maker and a regulatory path that are the company's own. Oxidized zirconium is Smith and Nephew's OXINIUM route, not a generic zirconium bar for every implant maker, and shipment, approval and share are unconfirmed.

Table 13: Benchmark medical applications, outlook to 2035

Outlook	Applications	Form
Option	Hip and knee implants where the company holds the path	Cobalt alloy and zirconium bar
Not a generic bar	Oxidized zirconium parts	Smith and Nephew OXINIUM route
Option, financed printer only	Printed implants	Cobalt chromium powder
Keep only if already profitable	Instruments and imaging	Stainless wire and nickel alloy strip
Stop	China implant tenders	Cobalt alloy bar

Sources: The application list is the VDM Metals portfolio used as a benchmark only, the China price cut is (National Healthcare Security Administration, 2022), and the outlook is the Chapter 8 verb rather than a Münvest qualification.

Stryker sold about USD 2.7 billion of knees and USD 1.9 billion of hips in 2025, and Zimmer Biomet sold about USD 3.3 billion and USD 2.1 billion, while Smith and Nephew reported revenue of about USD 6.2 billion (Smith & Nephew plc, 2026; Stryker Corporation, 2026; Zimmer Biomet Holdings, Inc., 2026). Those figures are OEM revenues, and for an alloy mill Carpenter medical sales of USD 224 million exclude surcharge, with the same book at USD 278 million including surcharge and down 21 percent (Carpenter Technology, 2026a). That Carpenter figure is the ceiling this chapter uses for the mill, because the OEM totals are implant maker revenue. Acerinox High Performance Alloys is named once if a cobalt chromium grade is contested, and its 47 thousand finished tonnes are not medical tonnes (Acerinox, 2026d).

What the company should do: the company should treat zirconium and cobalt implant bar as an option until an implant maker and a regulatory path are its own, and it should not approach a buyer on the strength of a catalogue. Powders stay inside the same financed programme gate used in energy and in aerospace. China implant tenders are on the stop list after the price cuts, and instrument and imaging business is harvested only where it is already profitable and less exposed to those implant prices.

6. Global Market Dynamics and Trade

Trade is the point at which the supply profiles become a geography of buyers and sellers. A country can export ore or primary metal and still import the plate, bar or tube a buyer specifies, so production leadership does not by itself create access to a qualified form. This

chapter uses UN Comtrade for defined HS codes and years and Eurostat PRODCOM for the European legs, and every figure below names the product, the year and the reporter. The quantity identity in section 4.1 is not repeated, because what this chapter adds is the unit value beside the tonne, the rounding band on bar, and the partner corridors for the codes the priorities ship in.

6.1 Product classifications

The nickel mill forms map to HS 7505 for bars, rods, profiles and wire, HS 7506 for plates, sheets, strip and foil, and HS 7507 for tubes and fittings. HS 7502 is unwrought nickel, which is outside the mill product identity and is where rotating billet is often declared. Adjacent codes cover cobalt, titanium, welding electrodes and aero engines, and the HS headings still mix pure metal with alloy inside one code, so the four digit split of HS 7506 into unalloyed sheet and nickel alloy sheet is not in the source log and no corridor below is called qualified alloy demand (United Nations Statistics Division, 2026c).

6.2 Unit values beside the nickel tonnes

Table 14 places a team estimate of euros per tonne beside the 2024 PRODCOM tonnes. EU flat sold production is about EUR 14,476 per tonne, German sold production is about EUR 8,788 per tonne, and EU flat imports are about EUR 47,824 per tonne, against about EUR 35,809 per finished tonne at Acerinox High Performance Alloys (Team, 2026i, 2026l, 2026m, 2026n). The gap is the reason 57.8 million kilograms stays an upper bound on a mixed code.

Table 14: Nickel mill product tonnes and unit values, 2024

Area and code	Sold production	Imports	Exports	Apparent consumption	Sold production, EUR per tonne
EU27 flat 24452200	80.9 million kg, estimated flag	14.4 million kg	37.5 million kg	57.8 million kg, mixed code ceiling	14,476
Germany flat 24452200	30.6 million kg	17.9 million kg	18.1 million kg	30.4 million kg	8,788
EU27 bar 24452100	2.5 million kg, rounded	about 5.6 million kg	about 2.70 million kg	5.38 million kg, band about 5.1 to 5.6	not used, cell is rounded
EU27 tube 24452300	21.0 million kg, round	4.7 million kg	16.5 million kg	9.2 million kg	not a capacity case

Sources: Apparent consumption (Eurostat, 2025a, 2025b, 2025c, 2025d). Unit values and the bar band are team estimates (Team, 2026f, 2026l, 2026n). German exports of 18,087,152 kilograms sit next to Comtrade HS 7506 exports of 17,775 tonnes (Eurostat, 2026a; United Nations, 2026).

Bar, rod and wire is the form where the published EU identity is a net import, and the net importer conclusion holds inside the rounding band (Team, 2026f). Exports on that row sit above the rounded production figure, which means stock or an estimate that is not a census. Flat products are a net export of a mixed code, so a flat seller faces a code the Union already exports, while a bar seller faces a logged net import that is still a rounded estimate, with United Kingdom supply outside the customs territory. There is no bar investment case until the rounding footnote, United Kingdom supply and HS 7502 are read against the client's own bar book, and that test is a comparison rather than a proposal to add capacity.

6.3 Corridors for the three priorities

Table 15 gives the reporter, the year, the net position and the leading partners for the codes tied to aerospace bar and billet, alloyed flat, welding wire, titanium, cobalt and engines. OECD intermediate use of non ferrous metals, USD 220 billion into electrical equipment and USD 109 billion into motor vehicles in 2022, stays an industry value and does not measure a corridor (OECD, 2025b, 2025c).

Table 15: Germany 2024 corridors, and the engine exporters in the same year

Reporter and code	Net position	Leading partners by value
Germany HS 7502 unwrought	Net import, 54.6 million kg in and 13.6 million kg out	Norway, Russian Federation, United Kingdom
Germany HS 7505 bar and wire	Nearly balanced on weight, 18.1 million kg out and 17.9 million kg in	Exports to the United Kingdom and the United States, imports from Austria
Germany HS 7506 flat	Net export of a mixed code, 17.8 million kg out and 5.9 million kg in	Austria, Italy, China
Germany HS 7507 tube	Net export, 3.1 million kg out and 2.4 million kg in	United States and Italy
Germany HS 8108 titanium	Net import, 26.6 million kg in and 18.4 million kg out	Russian Federation and the United States
Germany HS 8311 welding	Net import on weight, net export on value	Not nickel alloy wire alone
United Kingdom, France, United States HS 8411	Exports of USD 39.8 billion, USD 22.0 billion and USD 14.1 billion	Includes parts, not complete engines only

Sources: Bilateral extracts (United Nations Statistics Division, 2026a, 2026b, 2026c, 2026d, 2026f, 2026g, 2026h, 2026i). Cobalt HS 8105 is a small German net export of a mixed code and is not repeated as a superalloy balance (United Nations Statistics Division, 2026e).

Qualified aerospace product for a wrought mill is billet, bar, sheet and weld wire, read against HS 7505 and HS 7502, not against the flat balance. Corrosion resistant plate is alloyed flat, read against the alloy split once it exists. Expansion alloys, Crofer and much of the soft magnetic book are chapter 72, Konstantan is chapter 74, and Nickel 201 strip is unalloyed nickel at a different price from Alloy C-276 plate. One flat balance cannot size all three priorities, which is why the scenarios in the next chapter test application demand and do not convert these corridors into a company tonne.

7. Market Scenarios to 2035

The 2035 outlook tests demand for the relevant applications under three cases, and it is not a forecast of industry revenue, company sales or alloy tonnes. The 2025 physical base for each priority is unknown, pending the shipment map, because no defensible share of the mixed flat code or of the rounded bar code can be stated (Eurostat, 2025b, 2025d). The indices are

therefore not multiplied by a tonne range in this chapter or in Chapter 8. Metal prices are held constant, and no company total is printed, because 2025 shipment weights were not available.

7.1 Market Model and Scenario Assumptions

For an industry and a scenario, the 2035 index equals 100 times (1 plus the annual rate) to the power 10. A company weighted index would multiply each industry index by that industry's 2025 shipment share. Those weights were not supplied, and the rates are author choices, so each line below names the driver, the year and the adjustment, using figures already in the report.

Automotive uses the team estimate that cars without a plug fall from 58.2 million to 42.9 million, and a car with a plug can still carry an engine. The downside of 2 percent a year is the faster loss of diesel only parts. The base of 1 percent a year is that car path with turbo content slowing the decline. The upside of 0.5 percent a year is the hybrid path, so the upside is not a flat zero (Team, 2026e). The October 2026 committee vote stays out of the model because it is not a logged value row.

Oil and gas uses the 2025 upstream level of just under USD 570 billion as context for replacement, not as a 2035 spending path (International Energy Agency, 2025g). The downside of 1 percent a year is slower replacement, the base of 0.5 percent a year is replacement that holds, and the upside of 1.5 percent a year is that replacement plus funded gas projects. None of these rates is the USD 65 billion Current Policies gap.

Energy uses committed low emissions hydrogen of 4.3 million tonnes a year by 2030 as the bankable class (International Energy Agency, 2026j). The downside of zero is that committed class with no extra alloy, the base of 3 percent a year allows some replacement equipment on committed projects, and the upside of 6 percent a year is the conditional case above 6 million tonnes if further projects are decided, which is not a stock already committed. Electricity growth is not applied to alloy tonnes in any of the three cases.

Aerospace separates new engine hardware from shop visits, and LEAP deliveries rose 28 percent in 2025, and shop visits are the larger OEM profit pool, while the mill sees weld wire, some replacement hardware and the original billet (GE Aerospace, 2026c; Safran, 2026a, 2026b). The fleet path of about 3.4 percent a year is a team interpolation from 23,310 to 45,550 and is not the alloy rate (Team, 2026j). The base of 3 percent a year sits near that interpolation and is still not a tonne forecast. The upside of 4.5 percent a year assumes both hardware and shop visits stay tight. The downside of 1 percent a year still compounds because ceramic matrix composites take the hottest static parts and do not retire the disc. Growth in that downside stops if an approval is missed, if substitution reaches the disc, or if both occur. The index is not a forecast of alloy tonnes, and 11,499 tonnes of aviation cobalt are not scaled into a nickel indicator (Cobalt Institute, 2026c).

Chemicals uses replacement rather than EU chemical sales, so the downside of 0.5 percent a year is the weak 2025 chemical year persisting (Acerinox, 2026h). The base of 1.5 percent a year is replacement of named grades. The upside of 3 percent a year is funded export projects through fabricators.

Electronics uses component alloys rather than the 26.2 percent chip growth (World Semiconductor Trade Statistics, 2026). The downside is a zero rate because chip growth is not the alloy rate, the base of 2.5 percent a year is repeat orders for controlled properties, and the upside of 4.5 percent a year is that book with a wider set of named stampers. Chapter 72 tonnes are not taken from the nickel flat code in any of those cases.

Medical procedures compound at 3.36 percent a year on the constant set from 2013 to 2023, while the device illustration runs at 5.3 percent a year from 2023 to 2035 (OECD, 2025a). The application row here is 2 percent a year from 2025 in the base, which is a slower author choice because the mill ceiling is the Carpenter medical book and China tenders are stopped (Carpenter Technology, 2026a). The downside is 0.5 percent a year and the upside is 3.5 percent a year. The two windows are different, and neither window is converted into a bar tonne.

7.2 Results by Industry, Product, and Region

Table 16 reports the 2035 indices, with 2025 set to 100. The values are calculated and rounded. Appendix A2 shows all 21 unrounded results. No chart is added, because the table already carries the comparison and a figure would repeat it.

Table 16: Application demand indices for 2035 (2025 = 100)

Industry	Downside	Base	Upside	Annual rate, downside, base, upside	Why the downside still moves
Automotive	82	90	105	-2 percent, -1 percent, +0.5 percent	Diesel only parts fall faster than the hybrid engine
Oil and gas	90	105	116	-1 percent, +0.5 percent, +1.5 percent	Replacement can slow without a spending total
Energy	100	134	179	0 percent, +3 percent, +6 percent	Unfunded hydrogen adds no alloy
Aerospace	110	134	155	+1 percent, +3 percent, +4.5 percent	Discs remain if only the hottest static parts are substituted
Chemicals	95	116	134	-0.5 percent, +1.5 percent, +3 percent	Named grade replacement can shrink in a weak year
Electronics	100	128	155	0 percent, +2.5 percent, +4.5 percent	Chip sales are not the alloy rate
Medical	105	122	141	+0.5 percent, +2 percent, +3.5 percent	Procedures are not mill tonnes

Sources: The indices are a team estimate from the annual rates in this chapter, the automotive upside replaces a flat zero so that the hybrid path in section 5.1 is visible (Team, 2026e), and Appendix A2 shows the unrounded values, which are not tonnes.

Products stay tied to the codes in Chapter 6, and regions stay qualitative because shipment shares are unavailable. In the base case the priorities are aerospace billet, approved sheet and repair wire in Europe and North America, chemical plate and welding products through fabricators, and precision strip and wire for electrical components where the code is the one the grade ships in. Energy products follow committed projects rather than the announced pipeline, and the 2025 tonne range for each priority is unknown, pending the shipment map, so Chapter 8 does not convert Table 16 into a capacity tonne. Chapter 8 turns the same ranking into a sequence of commercial actions.

8. Strategic Recommendations

The ranking follows Table 2, the flat and bar positions in Chapter 6, and the four verbs already used in Chapter 5. Aerospace billet, approved sheet and repair wire come first, corrosion resistant plate and welding products come second only where a named grade is imported or replaces a specified supplier, and precision electrical strip and wire come third only in the code that grade ships in. Oil and gas and profitable automotive accounts are kept, electrolysers, powders and medical implant bar stay options, and undifferentiated flat tonnes, diesel only parts, China implant tenders and unfunded programmes are stopped. No aggregate market value is stated, and 57.8 million kilograms is not treated as room under the 47 thousand finished tonnes.

8.1 Priority Products, Industries, Regions, and Customers

Three sentences are enough for a plant director to use without a second reading of Chapter 4. Bar, rod and wire is a logged EU net import, with apparent consumption of 5,380,347 kilograms inside a band of about 5.1 million to 5.6 million kilograms, and the commercial test in months 0 to 3 is the client's own bar book against that position rather than a proposal to add bar capacity (Eurostat, 2025b; Team, 2026f). Flat products are a net export of a mixed code, so chemical plate and electrical strip are pursued only where a named grade is imported or a replacement order displaces a specified supplier, and undifferentiated flat tonnes are refused (Eurostat, 2025d). The 57.8 million kilograms is the ceiling of the code, the qualified alloy slice is smaller and still unmeasured, and there is no bar investment case until the rounding footnote, United Kingdom supply and HS 7502 are in the comparison (United Nations Statistics Division, 2026a).

The ranking has to sit on the code the product ships in, so qualified aerospace product for a wrought mill is billet, bar, sheet and weld wire, read against HS 7505 and HS 7502, not against the flat balance. Corrosion resistant plate is alloyed flat, read against the alloy split of HS 7506 once that split exists. Expansion alloys 29-18 and 42, Crofer 22, Aluchrom and much of the soft magnetic book are iron predominant and sit in chapter 72, Konstantan and other copper nickel resistance alloys sit in chapter 74, and Nickel 201 electrolyser strip is unalloyed nickel at a different price and a different buyer from Alloy C-276 plate. One flat balance cannot size all three priorities, and in this industry engine means the aero engine and the industrial gas turbine, so the words declining engine only applications are reserved for automotive engine and exhaust parts such as turbochargers, valves, plugs and exhaust. Haynes, inside Acerinox High Performance Alloys, is the book facing aerospace and industrial gas turbines, and VDM is the book facing the weak chemical and oil and gas year, with the two still counted once (Acerinox, 2026d, 2026g).

Table 17: The eight criteria in words, with no composite score

Criterion	Aerospace bar, sheet and wire	Chemical plate and weld products	Electrical strip and wire	Keep, and do not fund
Market attractiveness	High, because shop visits are the larger OEM pool	Medium, in a weak 2025 chemical year	Medium, and much of the book is chapter 72	Oil and gas is replacement, automotive is a shrinking engine set, medical OEM revenue is not the mill
Growth to 2035	Medium on the index, and the index is not tonnes	Medium on replacement, not on chemical sales	Medium on components, not on chip sales	Automotive falls, medical procedures are not bar, unfunded hydrogen is zero
Strategic fit	High only for a grade the mill already makes	High for plate and weld products on a named duty	Medium where the mill already finishes the strip	Low wherever the approval is not the company's
Competitive intensity	High: ATI about 68 percent and Carpenter about 65 percent of sales in aerospace and defence, and Acerinox finished 47 thousand tonnes in 2025	High on commodity 625, including Chinese mills, and a different contest on a patented Haynes grade	Medium: Aperam Imphy, Proterial, Nippon Yakin, VDM inside Acerinox	Alleima, Special Metals, Nippon Steel and Tubacex on sour service
Technology attractiveness	High for approved wrought forms, low for powders	Medium, on known grades	Medium, on property control	Low for a generic zirconium bar
Customer attractiveness	High, and concentrated in five engine makers	Medium, through fabricators	Medium, through stampers that pay for the property	Medical OEMs keep the margin
Entry barriers	High: a multi year rotating path	Lower: a fabricator trial can close inside a year, while	Medium: a sample and a property test	High for an implant path the company does not hold

Criterion	Aerospace bar, sheet and wire	Chemical plate and weld products	Electrical strip and wire	Keep, and do not fund
		sour service stays multi year		
Differentiation potential	Medium if already approved, low as a new disc source	Medium on a named grade against commodity 625	Medium on consistency	Low for stainless that already meets the duty

Sources: ATI and Carpenter shares and the 47 thousand tonnes are (Acerinox, 2026d; ATI Inc., 2026a; Carpenter Technology, 2026c). The other cells are the team reading of Chapters 5 and 6. No composite score is published because 2025 shipment shares were not available. Weights stay in the project model.

Table 18: Where effort stops, or where an account is only harvested

Application	Form	Why effort stops or why the account is only harvested	2025 revenue
Glow plugs and the diesel only set	Wire and bar	Volume falls faster than the rest of the engine set	Unknown until the month 0 to 3 map
Undifferentiated stainless	Flat product	The duty is already met by stainless	Unknown until the month 0 to 3 map
China implant tenders	Bar	Prices were cut by 82 percent	Unknown until the month 0 to 3 map
Unfunded hydrogen	Strip	No financed buyer	Unknown until the month 0 to 3 map
Powders without a printer programme	Powder	No financed programme in energy or in aerospace	Unknown until the month 0 to 3 map
Oil and gas without the company's own approval	Bar and plate	A catalogue listing does not travel	Unknown until the month 0 to 3 map
Automotive accounts that are already profitable	Engine and exhaust forms	Harvest only, and magnetic sensors sit in section 5.6	Unknown until the month 0 to 3 map

Sources: The stop list is the same verb set as Chapter 5, the China price cut is (National Healthcare Security Administration, 2022), and revenue by grade is an internal fact that is not invented here.

8.2 Recommended Actions and Implementation Roadmap

Months 0 to 3 test the European bar position from Chapter 6 against the client's bar book, and they take glow plugs off the growth list now. Months 3 to 12 cap the live qualification list at five accounts and cap sample spend at EUR 200,000 per account until a quantity is agreed, and both caps are team planning choices rather than market facts (Team, 2026g, 2026h). Months 12 to 24 release finishing capacity only against a repeat order and only where contribution per bottleneck hour still clears the Chapter 7 downside. The hurdle rate stays the company's, and the memo shows the number the company uses rather than a rate invented here.

Two clocks run at the same time through the full 24 months, so fabricated chemical equipment can be trialled inside a year, while rotating parts and sour service casing stay on a multi year approved supplier path, and a 3 to 12 month trial does not cover a disc. The kill rules sit in that same sequence, so no second sample is cut without a quantity, no finishing capital is released without a repeat order and a downside case that clears the company's hurdle, and no powder or electrolyser spend is released without a financed programme.

The invoice has to be written the way the buyer is charged. Nickel alloy plate is a conversion price plus a monthly alloy adjustment on the nickel, molybdenum, chromium, cobalt, tungsten and niobium in the grade, plus extras for width, testing, positive material identification and non destructive testing. The stainless invoice, a base price plus an alloy surcharge, is a different document. The first 90 day map has to say who carries nickel, molybdenum and cobalt between order and invoice, and whether scrap returns to the same mill, and it has to ask for contribution per bottleneck hour in the company's own grade and form codes. The melt routes stay distinct: vacuum induction plus vacuum arc remelting for rotating grades, vacuum induction plus electroslag remelting for corrosion plate, and a stainless argon oxygen route for Crofer and for the lower nickel stainless grades. Appendix A4 remains the list of internal facts still required, and it is not a reason to postpone the external choices already made from the trade result and the stop list.

9. Conclusion

The choice is the same one Chapter 1 stated, and it is repeated here only so the close matches the opening. Qualify aerospace billet, approved sheet and repair wire, and qualify chemical plate and welding products only where a named grade clears the capacity rule. Qualify electrical strip and wire only in the code that grade ships in. Keep oil and gas where the company's own approval supports the margin, and harvest automotive accounts that are already profitable. Treat electrolysers, powders and medical implant bar as options against a financed programme or a regulatory path the company holds. Stop glow plugs and the diesel only set, undifferentiated stainless, China implant tenders, and any powder or electrolyser spend without a financed buyer.

The scale behind that choice is Table 2, not a specialty reading of one trade code. Mine production and primary use are millions of tonnes of nickel, stainless melt is tens of millions of tonnes of steel, and EU flat apparent consumption of 57.8 million kilograms in 2024 is the ceiling of a mixed code that still joins pure nickel and nickel alloy (Eurostat, 2025d; U.S. Geological Survey, 2026s; worldstainless, 2026a). That ceiling is not qualified alloy demand and it is not room under the Acerinox finish figure. Acerinox High Performance Alloys finished 47 thousand tonnes in 2025, worldwide, with VDM and Haynes counted once, and that figure is output rather than a European capacity ceiling (Acerinox, 2026d). Flat product is a net export of the mixed code, and bar is a logged net import that is still a rounded estimate, so the bar test is the client's own book and not a case to add capacity (Eurostat, 2025b; Team, 2026f).

The 2035 indices stay in Chapter 7 and are not converted into tonnes, because the 2025 share of each priority inside the relevant code is unknown until the shipment map. The first 24 months use the trade result and the stop list, cap the qualification list, and release finishing capacity only against a repeat order that still clears the downside. Internal sales, margins, approvals and nameplate by plant are still required before those gates become a capital request, and they are not a reason to reopen the ranking.

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Appendices: Data, Sources, Calculations, and Assumptions

A1 Definitions and evidence

Mt means million tonnes, GW means gigawatts, and EBITDA means earnings before interest, tax, depreciation and amortisation. Industry revenue, mined metal, melting output and finished alloy shipments have different boundaries, and each figure keeps its original currency and observation year. No currency conversion or inflation adjustment is applied to the scenario indices. Web sources for the newly completed sections were consulted on 28 October 2026, and earlier observation years are used where the text states them.

Apparent consumption equals sold production plus imports minus exports for one PRODCOM code, one area and one year (Team, 2026d). The logged results are EU nickel flat products at

57.8 million kilograms, German nickel flat products at 30.4 million kilograms, EU nickel bar, rod and wire at 5,380,347 kilograms, and EU nickel tube, pipes and fittings at 9.2 million kilograms, all for 2024 (Eurostat, 2025a, 2025b, 2025c, 2025d). The flat result is an upper bound on a mixed code, the EU flat sold production leg is estimated, EU trade in the dataset is extra EU trade, and the identity is not end use alloy demand and not stainless melt. Unit values are team estimates: about EUR 14,476 per tonne of EU flat sold production, about EUR 8,788 per tonne of German flat sold production, and about EUR 47,824 per tonne of EU flat imports (Team, 2026l, 2026m, 2026n). The bar band of about 5.1 million to 5.6 million kilograms is the rounding range around the published point, and the net importer conclusion holds inside it (Team, 2026f). German flat sold production rose from 15,870 tonnes in 2019 to 36,477 tonnes in 2020 while value stayed near EUR 177 million to EUR 191 million (Team, 2026k).

A2 Reproducible scenario calculations

The base year 2025 is set to 100, the horizon is 10 years, and the index equals 100 times (1 plus the annual rate) to the power 10. The unrounded values are the ones below, and Table 16 rounds each of them to the nearest integer. Prices are held constant, and no shipment weight is applied, so there is no company total in this appendix. The automotive upside rate is 0.5 percent a year, a team estimate that replaces a flat zero (Team, 2026e).

Industry	Downside, unrounded	Base, unrounded	Upside, unrounded
Automotive	81.71	90.44	105.11
Oil and gas	90.44	105.11	116.05
Energy and environmental technology	100.00	134.39	179.08
Aerospace	110.46	134.39	155.30
Chemicals	95.11	116.05	134.39
Electronics and electrical engineering	100.00	128.01	155.30
Medical technology	105.11	121.90	141.06

Sources: The twenty one values are a team estimate from the rates in Chapter 7, the aerospace downside still compounds because the hottest static parts can be substituted without retiring the disc, and these values are not tonnes.

The 2025 tonne range for aerospace bar and wire, for chemical plate, and for electrical strip is unknown, pending the shipment map. No share of PRODCOM 24452200 or 24452100 is

applied to the indices. The aviation cobalt illustration of about 16,100 tonnes holds 2025 intensity constant with the fleet interpolation and is not a nickel indicator (Team, 2026a).

A3 Interpretation of existing sector estimates

The automotive calculation groups battery electric and plug in hybrid cars together, and cars with a plug can still contain combustion engine components. That is why the automotive upside is not a flat zero, and fleet growth is not a direct measure of new engine alloy purchases, and holding cobalt intensity constant is a simplifying assumption rather than a nickel forecast. The medical device value extrapolation applies a broader bio health growth proxy from 2023 to 2035 at 5.3 percent a year and is not the Chapter 7 medical rate, which runs from 2025. Upstream oil and gas spending, chemical sales, device sales and aviation cobalt stay illustrations of activity. They are not alloy demand, and they are not an addressable market total.

A4 Information needed before investment

The external choices in Chapters 6 and 8 do not wait for this list. The first gate uses the logged bar position, the mixed code ceiling and the stop list. Before a capital request, the company still needs shipments and net of metal sales by grade, form, application, region, margin and approval, contribution per bottleneck hour, and the nameplate by plant and by route. The melt footnote should be quoted before anyone adds the 83 thousand tonnes of 2025 melt to stainless melt, because a vacuum induction heat and a later remelt of the same metal would double count. The 2028 furnace and finishing steps are not inside the 2025 tonnes. VDM and Haynes are counted once inside Acerinox, and the alloy split of HS 7506, United Kingdom bar supply, and HS 7502 billet declarations are required before any bar capacity case. Tube stays off the capacity list until a route the company owns is distinguished from the rounded PRODCOM tube code. Scenario assumptions should be replaced with customer quantities when those quantities exist, and the hurdle rate used in the memo should be the company's own.